

might do well to examine the maintenance ratio (*i.e.*, the percentage of gross revenues expended on upkeep of way and rolling stock) in order to make sure that it is not suspiciously below standard. Unfortunately it is difficult to determine with any degree of assurance just what should be considered a standard maintenance rate for different groups of carriers. Prior to 1931 a figure of about 33% of gross operating revenues was so generally and consistently reported that it undoubtedly could be considered a norm, any wide deviation from which deserved special study.⁴ Since 1930, however, there has been a moderate decline in this percentage figure concurrently with a major shrinkage in the gross operating revenues against which it is computed. As a result, actual dollar expenditures for maintenance have been cut nearly in half. (Somewhat surprisingly, the maintenance-of-way outlays in dollars—which presumably are not so subject to curtailment on account of smaller traffic—suffered a decline of 51% in 1933–1937 as against 1926–1930, whereas maintenance of equipment costs were reduced by 39%.)

On the other side must be set the undoubted improvement in the technology of maintenance as shown in the use of more efficient methods and more durable materials.⁵ The cost of maintaining railroad property in adequate condition is now substantially less than it was prior to 1931. But how much less we cannot say with assurance; hence the difficulty of determining whether the average ratio of about 30¹/₂% on the reduced gross of 1933–1937 (shown by all Class I Railroads) is sufficient to reassure the bond buyer against the existence of undermaintenance. Our judgment leans to the view that this figure is rather low⁶ and that a somewhat higher ratio—say 32%—might better be taken as the investor's norm.

⁴ Geographical differences, formerly productive of rather wide variations in the customary maintenance ratio, were not of great importance in the years 1926–1930. See material on this point and others relating to railroad maintenance in Appendix Note 23.

⁵ Many detailed examples on this point are given in an address of L. A. Downs, president of the Illinois Central Railroad, delivered Dec. 3, 1936, and reprinted by the Association of American Railroads.

⁶ This conclusion is supported by the replies of the railroads themselves to a circular of the Interstate Commerce Commission, dated Dec. 12, 1938, in which they estimated that a total of \$283,800,000 of deferred maintenance existed on their lines at the end of 1938. The replies generally distinguished between maintenance sufficient for safe and economical operation, which they contended had been performed, and maintenance necessary to bring the property to a satisfactory engineering standard. (See summary of return on Statistical Series Circular 26, published as Statement 3911 by the Bureau of Statistics of the Interstate Commerce Commission, March 1939.)